RETAIL SALES & PROFIABILITY ANALYSIS DASHBOARD & REPORT

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RETAIL SALES & PROFIABILITY ANALYSIS DASHBOARD & REPORT – R & Tableau

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Tools used: RStudio, Tableau.

Project Overview:

In this project, I use R to analyze and clean data from an online sports store. The goal is to answer key business questions and extract insights on revenue, profit, customer ratings, and geographic trends.

Additionally, I created an interactive Tableau dashboard to gain a deeper understanding of the data. You can explore it here:
https://public.tableau.com/app/profile/sebastian.ortuno.barrero/viz/SportsSalesCustomerAnalysis/RatingsDashboard?publish=y

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Data Cleaning and Business Requirements:

1) Convert 'date' column (in text format) to a proper DATE type and store in 'Date_New'.

2) KPIs: total revenue, profit, number of orders, profit margin.

3) KPIs by sport: revenue, profit, orders, profit margin.

4) Customer ratings: number, the percentage of ratings the company got from all the orders, average rating.

5) Ratings distribution: number of orders by rating, revenue by rating, profit by rating, and profit margin by rating.

6) Revenue, profit, and profit margin by State.

7) Monthly profit trends and comparison with previous month.

8) Monthly revenue trends and comparison with previous month.

Libraries:

Before we start with the business requirements, I load the libraries needed for this project.

```r
library(openxlsx)
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.1

## Warning: package 'ggplot2' was built under R version 4.5.1

## Warning: package 'dplyr' was built under R version 4.5.2

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(dplyr)
library(lubridate)
library(ggplot2)
library(knitr)
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```r
library(scales)
```

Load and view the dataset

After loading the data, we begin our analysis by using the `head()` and `glimpse()` functions to explore the structure and content of the dataset.

There are two tables: **orders** and **customers**.

- The **orders** table contains 8 columns and 2,847 rows. The columns are: `order_id`, `customer_id`, `sport`, `revenue`, `profit`, `shipping_cost`, `rating`, and `date`.
- The **customers** table contains 4 columns and 2,847 rows. The columns are: `customer_id`, `full_name`, `email`, and `state`.

```r
orders<-read.csv("C:/Users/sebas/OneDrive/Documents/NEC MASTERS/Projects Portfolio/Projects Portfolio/R

customer<-read.csv("C:/Users/sebas/OneDrive/Documents/NEC MASTERS/Projects Portfolio/Projects Portfolio/

head(orders)
```

```
##       date order_id customer_id      sport revenue profit shipping_cost rating
## 1 1/1/2022    10001      102278   Baseball  183.60  97.29             0     NA
## 2 1/1/2022    10002      102279 Basketball  185.76 103.40             0     NA
## 3 1/1/2022    10003      102280 Basketball  128.16  66.27             0     NA
## 4 1/1/2022    10004      102281     Hockey   45.62  15.46             7     NA
## 5 1/1/2022    10005      102282   Football  106.30  21.75             0     NA
## 6 1/1/2022    10006      102283   Football   58.11  12.08             0      3
```

```r
head(customer)
```

```
##   customer_id     full_name                    email        State
## 1      102278   Alica Reary areary0@sciencedaily.com      Florida
## 2      102279  Delmor Rubin      drubin1@yahoo.co.jp      Indiana
## 3      102280   Joanie Hoyt     jhoyt2@bloglovin.com Pennsylvania
## 4      102281 Madelena Boat  mboat3@surveymonkey.com       Nevada
## 5      102282 Sayers Patkin       spatkin4@sogou.com     New York
## 6      102283  Merwyn Stout       mstout5@sfgate.com     Michigan
```

```r
glimpse(orders)
```

```
## Rows: 2,847
## Columns: 8
## $ date          <chr> "1/1/2022", "1/1/2022", "1/1/2022", "1/1/2022", "1/1/202~
## $ order_id      <int> 10001, 10002, 10003, 10004, 10005, 10006, 10007, 10008, ~
## $ customer_id   <int> 102278, 102279, 102280, 102281, 102282, 102283, 102284, ~
## $ sport         <chr> "Baseball", "Basketball", "Basketball", "Hockey", "Footb~
## $ revenue       <dbl> 183.60, 185.76, 128.16, 45.62, 106.30, 58.11, 214.04, 37~
## $ profit        <dbl> 97.29, 103.40, 66.27, 15.46, 21.75, 12.08, 70.41, 2.55, ~
## $ shipping_cost <int> 0, 0, 0, 7, 0, 0, 0, 7, 0, 0, 7, 0, 0, 0, 0, 0, 0, 7, 7,~
## $ rating        <int> NA, NA, NA, NA, NA, 3, 2, NA, 2, NA, 5, 2, 2, NA, NA, 2,~
```

```r
glimpse(customer)
```

```
## Rows: 2,847
## Columns: 4
## $ customer_id <int> 102278, 102279, 102280, 102281, 102282, 102283, 102284, 10~
## $ full_name   <chr> "Alica Reary", "Delmor Rubin", "Joanie Hoyt", "Madelena Bo~
## $ email       <chr> "areary0@sciencedaily.com", "drubin1@yahoo.co.jp", "jhoyt2~
## $ State       <chr> "Florida", "Indiana", "Pennsylvania", "Nevada", "New York"~
```

Convert 'date' column (in text format) to a proper DATE type and store in 'New_date'.

```r
# Convert the 'date' column to a Date type and create a new column 'New_date'
orders<-orders%>%mutate(New_date=as_date(date, format="%m/%d/%Y"))
# Select and reorder columns, replacing 'date' with the new 'New_date' column
orders<-orders%>%select("order_id","customer_id","sport","revenue","profit","shipping_cost",
                        "rating","New_date")
```

1) KPIs: total revenue, profit, number of orders, profit margin.

```r
KPI<-orders%>%  # Start with the orders table
  summarize(                    # Use summarize to calculate key metrics
    Total_Revenue=sum(revenue),#Calculate revenue
    Total_Profit=sum(profit),  # Calculate profit
    N_orders=n_distinct(order_id),# calculate number of orders orders using summarize
    Profit_Margin=round((Total_Profit/Total_Revenue)*100,2)) #Calculate the profit margin

kable(KPI)
```

Total_Revenue	Total_Profit	N_orders	Profit_Margin
459418.4	284821.9	2847	62

- **Total Revenue:** $459,418.40
  The store generated nearly half a million in total sales — strong revenue.

- **Total Profit:** $284,821.90
  Profit makes up a significant portion of revenue, indicating healthy operations.
- **Number of Orders:** 2,847
  On average, each order generates about $161.35 in revenue
  (459,418.4 ÷ 2,847)
- **Profit Margin:** 62%
  Very high margin — over half of every dollar earned is profit, which is excellent for retail.

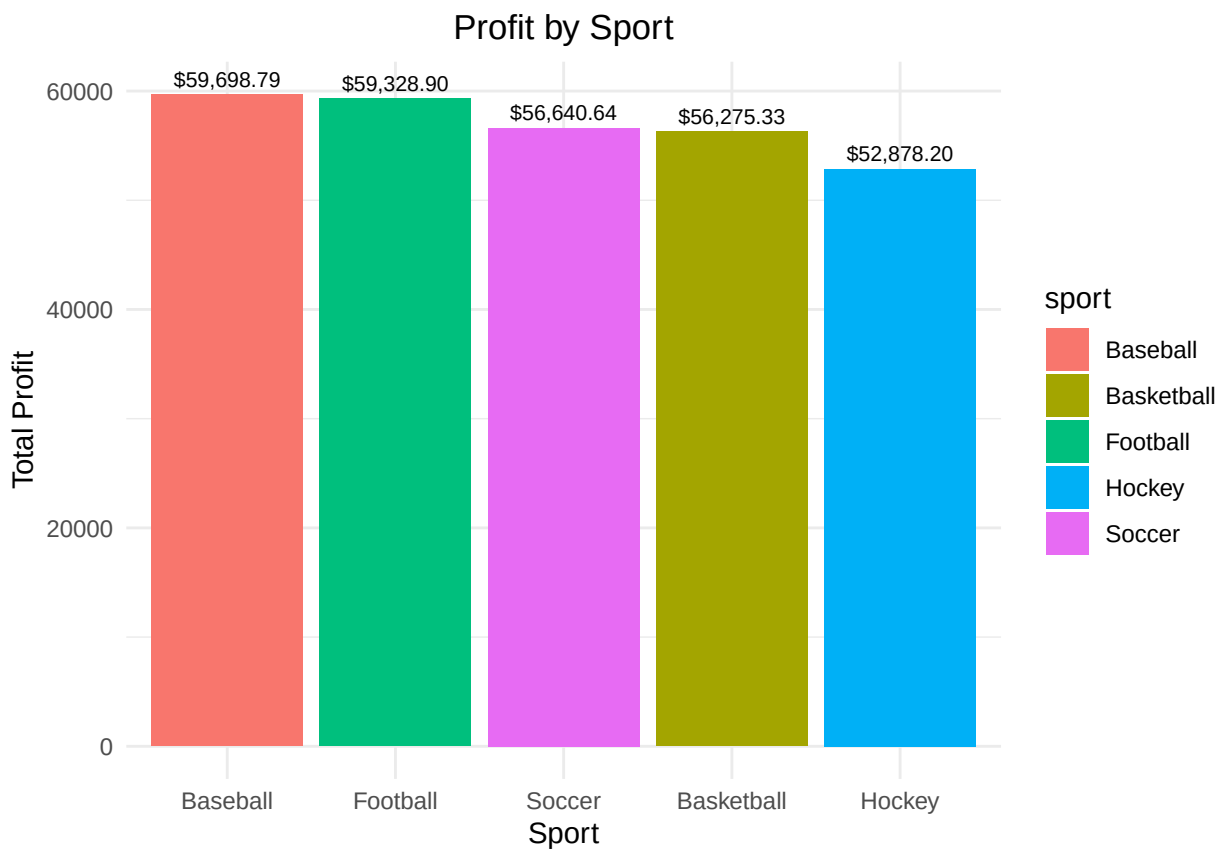
2) KPIs by sport: revenue, profit, orders, profit margin.

```
KPI_sports<-orders%>% # Start with the 'orders' dataset
  group_by(sport)%>% # Group data by 'sport' category
  summarise(Total_Revenue=sum(revenue), # Sum revenue for each sport
    Total_Profit=sum(profit),           # Sum profit for each sport
    Total_Orders=n_distinct(order_id),  # Count unique orders per sport
    Profit_Margin=round((Total_Profit/Total_Revenue)*100,2))%>%  # Calculate profit margin (%)
    arrange(desc(Total_Profit), desc(Total_Orders))  # Order results by highest profit, then orders

kable(KPI_sports) #Printing KPI_sports
```

sport	Total_Revenue	Total_Profit	Total_Orders	Profit_Margin
Baseball	95363.99	59698.79	565	62.60
Football	94768.32	59328.90	572	62.60
Soccer	90157.93	56640.64	561	62.82
Basketball	92116.23	56275.33	577	61.09
Hockey	87011.90	52878.20	572	60.77

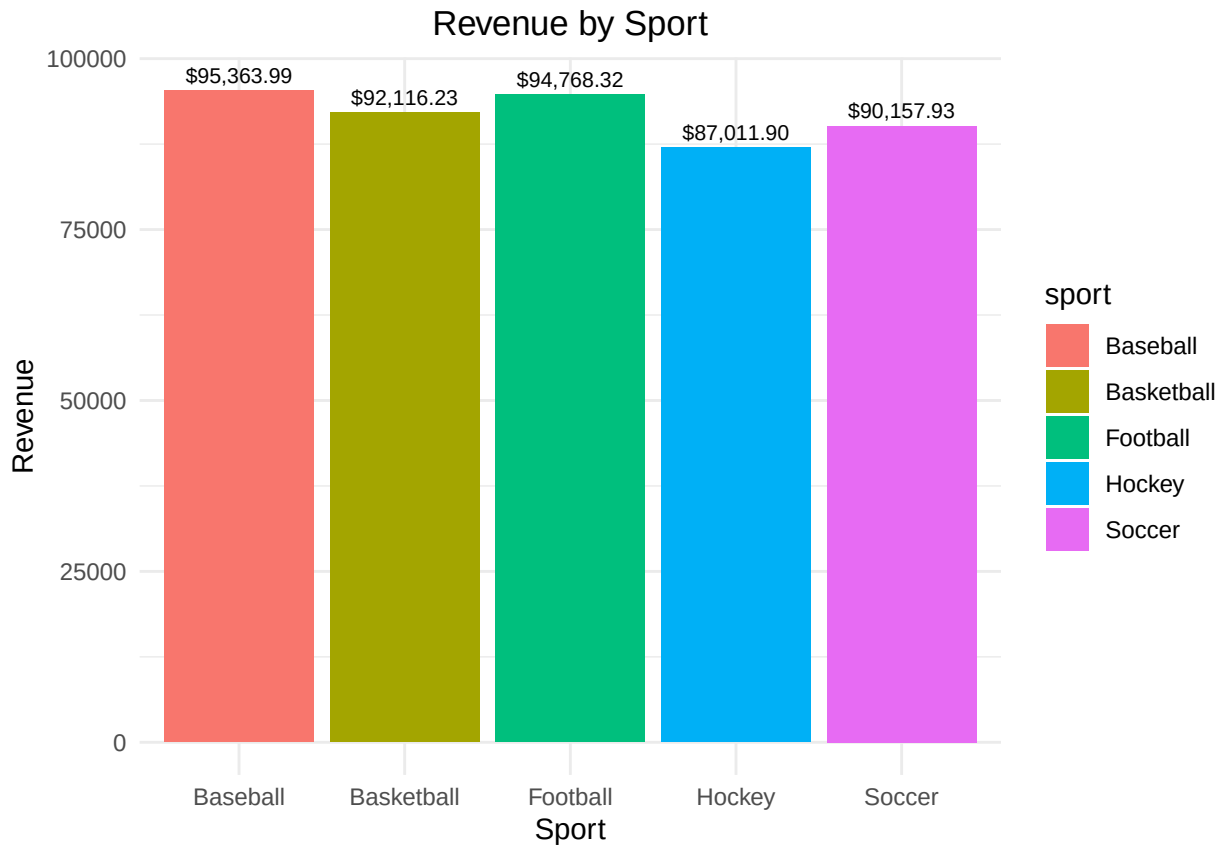
2.1 Profit by sports.

```
ggplot(KPI_sports, aes(x = reorder(sport, -Total_Profit), y = Total_Profit, fill = sport)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = dollar(Total_Profit)), vjust = -0.5, size = 2.7, ) +
  labs(title = "Profit by Sport", x = "Sport", y = "Total Profit") +
  theme_minimal()+
  theme(plot.title = element_text(hjust=0.5))
```



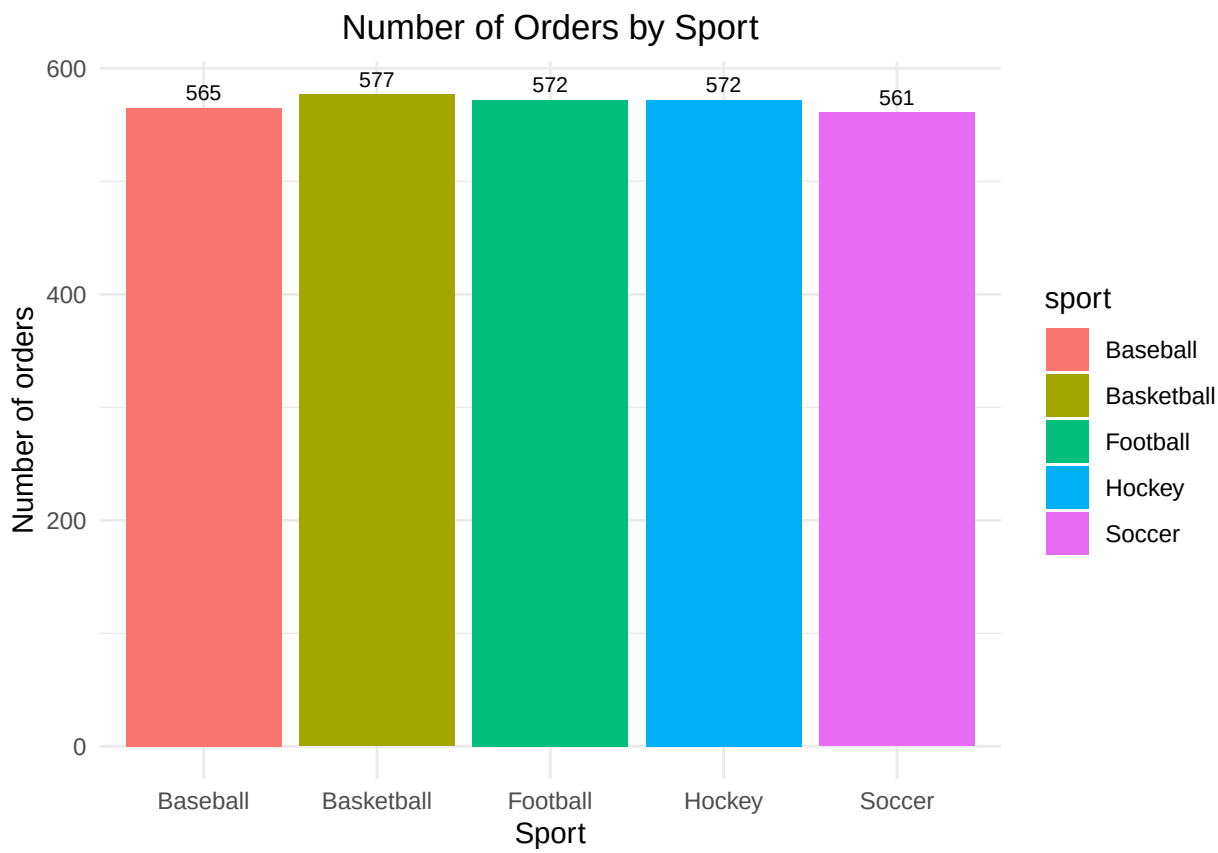
2.2 Revenue by Sport

```
ggplot(KPI_sports, aes(x=sport, y=Total_Revenue, fill=sport))+
  geom_bar(stat="Identity")+
  geom_text(aes(label = dollar(Total_Revenue)),vjust=-0.5, size= 2.8)+
  labs(title = "Revenue by Sport",
       x= "Sport",
       y="Revenue")+
  theme_minimal() +
  theme(plot.title = element_text(hjust=0.5))
```



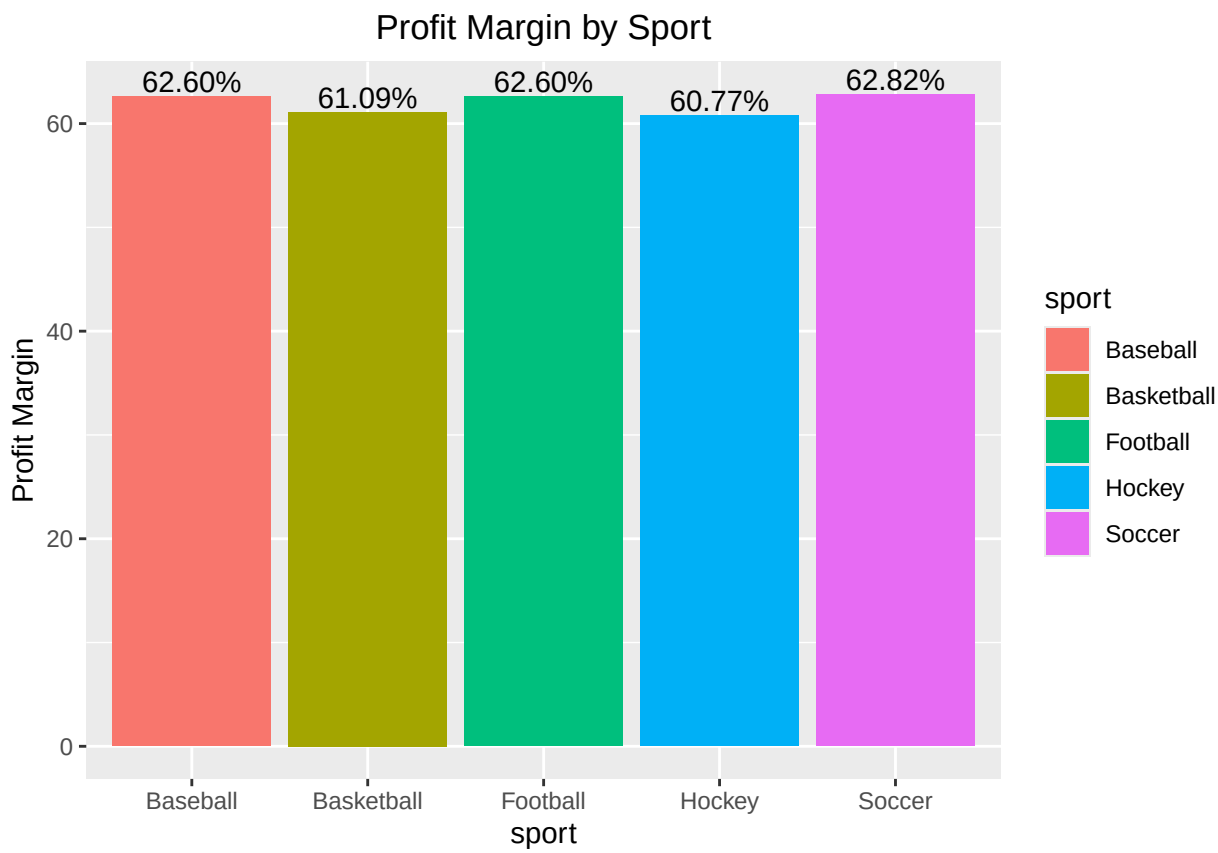
2.3 Number of orders by Sport.

```
ggplot(KPI_sports, aes(x=sport, y=Total_Orders, fill=sport))+
  geom_bar(stat="identity")+
  geom_text(aes(label=(Total_Orders)), vjust=-0.5, size=2.8)+
  labs(
    title= "Number of Orders by Sport",
    x= "Sport",
    y= "Number of orders")+
  theme_minimal()+
  theme(plot.title = element_text(hjust=0.5))
```



2.4 Profit Margin by Sport

```
ggplot(KPI_sports, aes(x=sport, y=Profit_Margin, fill=sport))+
  geom_bar(stat="identity")+
  geom_text(aes(label=percent(Profit_Margin/100), vjust=-0.2))+
  labs( title="Profit Margin by Sport",
        x="sport",
        y="Profit Margin")+
  theme(plot.title = element_text(hjust= 0.5))
```



2.5 Comments:

- All sports have profit margins above 60%, indicating strong overall profitability.
- **Soccer** has the highest profit margin, while **Basketball** leads in order volume.
- **Football** and **Baseball** show an excellent balance between high revenue and strong margins.
- **Hockey**, although slightly lower in margin, still performs well and could benefit from cost optimization strategies.
- The number of orders is relatively consistent across all sports categories.
- **Recommendation:**
  Consider reallocating marketing resources to **Hockey** and **Baseball** to boost volume, as they have strong performance potential. Additionally, analyze Basketball's pricing and cost structure to increase margins without sacrificing volume.

3) Customer ratings: number, percentage of ratings from all orders, average rating.

```
Customer_Ratings<-
    orders%>% # Start with the 'orders' table
    summarise(Average_Rating=round(mean(rating, na.rm=TRUE),2),  # Calculate the average rating (exclud
    Total_Rating=sum(!is.na(rating)),# Count the number of non-missing ratings
    Percentage_rating= round((Total_Rating/2847)*100,2))  # Calculate percentage of orders with ratings
```

```r
kable(Customer_Ratings)   # Display the summarized ratings info
```

Average_Rating	Total_Rating	Percentage_rating
3.13	1193	41.9

- **Average Rating:** 3.13
  The average customer rating across all orders is 3.13 out of 5, indicating a moderate level of customer satisfaction.
- **Total Ratings Submitted:** 1,193
  Out of 2,847 total orders, 1,193 included a customer rating.
- **Percentage of Orders with Ratings:** 41.9%
  Only 41.9% of customers submitted a rating, suggesting that the business may benefit from encouraging more feedback to better evaluate customer experience.
- **Recommendation:**
  Implement post-purchase incentives such as small discounts, loyalty points, or thank-you emails to encourage more customers to leave ratings. Increasing feedback volume will provide better insights for improving products and service.

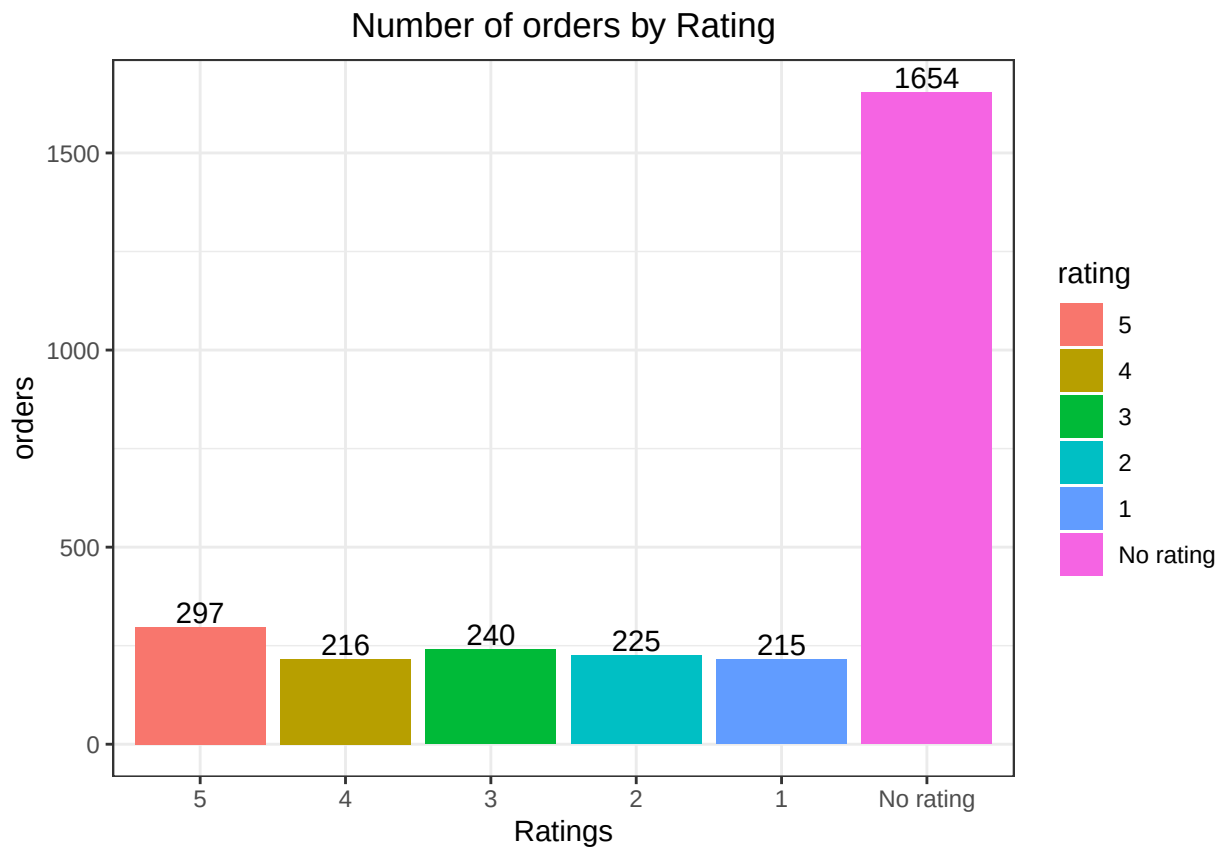
4) Ratings distribution: number of orders by rating, revenue by rating, profit by rating, and profit margin by rating.

```r
rating_distribution<-orders %>% # Use orders data
  mutate(
    rating = as.character(rating),  # First convert it to text.
    rating = replace_na(rating, "No rating"), # replace NA to "No rating provided"
    rating = factor(rating, levels = c("5", "4", "3", "2", "1", "No rating"))
  ) %>%
  group_by(rating) %>% summarise(orders_by_ratings=n(),     # Group by rating value
    revenue_by_rating=sum(revenue),                         # Count orders per rating
    profit_by_rating=sum(profit),                           # Sum revenue per rating
  profit_margin=round((profit_by_rating/revenue_by_rating)*100,2))%>% # Profit margin % per rating
  arrange(desc(rating))                                     # Sort by rating descending

kable(rating_distribution)                                          # Show results
```

rating	orders_by_ratings	revenue_by_rating	profit_by_rating	profit_margin
No rating	1654	290285.23	183760.92	63.30
1	215	28596.75	16339.88	57.14
2	225	31838.67	19250.57	60.46
3	240	38663.38	24209.14	62.62
4	216	29468.41	17303.72	58.72
5	297	40565.93	23957.63	59.06

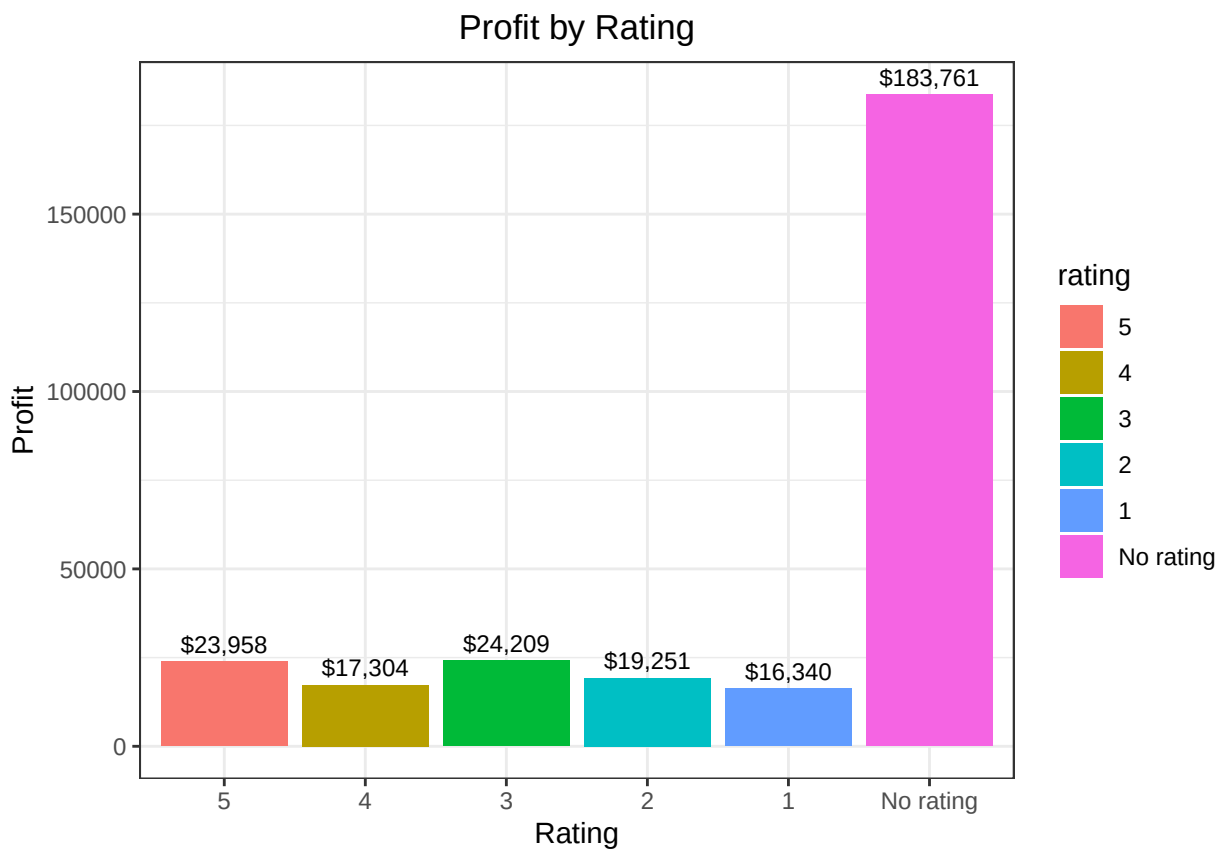
4.1 Orders by rating

```r
ggplot(rating_distribution, aes(x=rating, y=orders_by_ratings, fill=rating))+
  geom_bar(stat = "identity")+
  geom_text(aes(label = orders_by_ratings, vjust=-0.2))+
  theme_bw()+
  labs(title = "Number of orders by Rating",
       x="Ratings",
       y="orders")+
  theme(plot.title = element_text(hjust = 0.5))
```



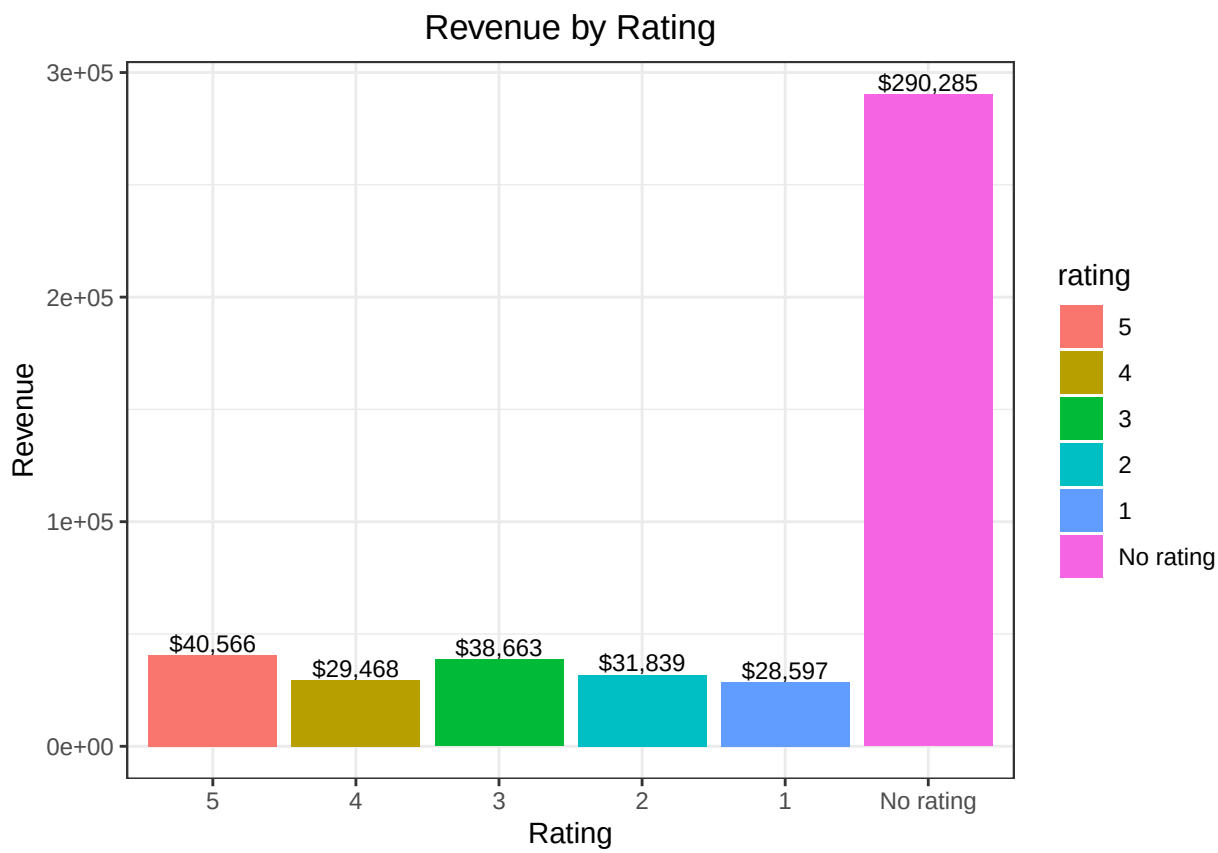
4.2 Profit by rating

```r
ggplot(rating_distribution, aes(x=rating, y=profit_by_rating, fill=rating)) +
  geom_bar (stat="identity")+
  geom_text(aes(label = dollar(profit_by_rating)), vjust=-0.5, size=3) +
  theme_bw()+
  labs(title= "Profit by Rating",
         x= "Rating",
         y="Profit")+
  theme(plot.title = element_text(hjust = 0.5))
```



4.3 Revenue by rating

```
ggplot(rating_distribution, aes(x=rating, y=revenue_by_rating, fill=rating))+
  geom_bar(stat="identity")+
  geom_text(aes(label=dollar(revenue_by_rating)), vjust=-0.2,size=3.2)+
  theme_bw()+
  labs(
    title="Revenue by Rating",
    x="Rating",
    y="Revenue"
  )+
  theme(plot.title =element_text(hjust = 0.5) )
```



4.4 Comments:

- **High number of unrated orders:**
  The majority of orders (1,654) have no rating, indicating many customers do not provide feedback.
- **Unrated orders generate the highest revenue and profit:**
  Despite lacking ratings, these orders account for the largest share of revenue ($290K+) and profit ($183K+).
- **Highest profit margin is from unrated orders (63.30%):**
  This is followed by rating 3 orders at 62.62%, and rating 2 at 60.46%.
- **Profit margins do not increase linearly with rating:**
  Rating 3 orders have higher profit margins than both ratings 4 and 5, which is unexpected.
- **Recommendation:**
  Investigate product categories and pricing strategies by rating to better understand these patterns and improve customer feedback collection.

5) Analyze revenue, profit, and profit margin by state.

5.1 Best profiability efficiency (Top 3)

```
profiability_efficiency<-inner_join(orders, customer, by="customer_id") %>% # Join orders with customer
  group_by(State) %>%                              # Group data by State
  summarise(Revenue_by_state= sum(revenue),        # Sum revenue per state
    profit_by_state = sum(profit),                 # Sum profit per state
```

```r
    profit_margin= round((profit_by_state/Revenue_by_state)*100,2)) %>%  # Calculate profit margin % pe
  mutate(rank_by_margin = as.integer(dense_rank(desc(profit_margin)))) %>%  # Rank states by descending
  arrange(rank_by_margin) %>%  # Order by rank (top margins first)
  filter(rank_by_margin<=3)     # Keep only top 3 states by profit margin

kable(profiability_efficiency)
```

State	Revenue_by_state	profit_by_state	profit_margin	rank_by_margin
Utah	5256.80	3656.88	69.56	1
Massachusetts	8664.92	6023.10	69.51	2
New Mexico	2996.67	2044.45	68.22	3

5.2 Highest revenue and profit (top 3).

```r
H_Revenue_Profit<-inner_join(orders, customer, by= "customer_id") %>%  # Join orders with customers by
      group_by(State) %>%      # Group data by State
      summarise(profit_state= sum(profit),  # Total profit per state
        revenue_state= sum(revenue),# Total revenue per state
        profit_margin= round( (profit_state/revenue_state)*100,2))%>% # Profit margin % per state
      arrange(desc(profit_state), desc(revenue_state)) %>% # Sort by highest profit, then revenue
      mutate(rank=row_number())%>% # Assign rank based on sorting
      filter(rank<=3)     # Keep top 3 states by profit

kable(H_Revenue_Profit)
```

State	profit_state	revenue_state	profit_margin	rank
California	34553.72	55470.47	62.29	1
Texas	32234.74	52305.67	61.63	2
Florida	22398.09	36250.92	61.79	3

5.3 least profitable and least revenue

```r
L_profitable_revenue<-inner_join(orders, customer, by="customer_id")%>% # Join orders and customers on
group_by(State) %>%                  # Group data by State
summarise(profit_state= sum(profit), # Sum of profit per state
  revenue_state= sum(revenue),       # Sum of revenue per state
  profit_margin= round( (profit_state/revenue_state)*100,2)) %>% # Profit margin % per state
arrange(profit_state, revenue_state)%>% #Sort by profit and revenue
mutate(rank= row_number()) %>%          # Assign rank based on sorted order
filter(rank<=3)                         # Keep only top 3 (lowest profit) states

kable(L_profitable_revenue)
```

State	profit_state	revenue_state	profit_margin	rank
Maine	16.72	91.10	18.35	1
Rhode Island	298.68	559.88	53.35	2
North Dakota	453.54	705.99	64.24	3

5.4 Smaller states like Delaware and New Hampshire show high margins despite lower total revenue.

```
small_States<-inner_join(orders, customer, by="customer_id")%>% # Join orders with customer data on cus
group_by(State) %>%                                # Group data by State
summarise(profit_state= sum(profit),               # Sum profit for each state
  revenue_state= sum(revenue),                     # Sum revenue for each state
  profit_margin= round( (profit_state/revenue_state)*100,2)) %>%   # Calculate profit margin (%) per st
arrange( desc(profit_margin)) %>%                  # Sort states by profit margin, highest first
mutate(Rank=rank(row_number()))%>%
filter(Rank==4 | Rank == 5)

kable(small_States)
```

State	profit_state	revenue_state	profit_margin	Rank
Delaware	1658.72	2446.65	67.80	4
New Hampshire	1012.14	1497.36	67.59	5

5.5 Comments:

- Utah, Massachusetts, and New Mexico have the best profit efficiency.
- California, Texas, and Florida rank highest in both Revenue and Profit, but not in margin.
- Maine and Rhode Island are at the bottom in all three metrics: least profitable and least revenue.
- Smaller states like Delaware and New Hampshire show high margins despite lower total revenue.
- **Recommendation:**
  Focus on improving operational efficiency in large-volume states like California, Texas, and Florida to enhance profit margins. Additionally, investigate strategies to boost sales in underperforming states like Maine and Rhode Island, potentially through targeted promotions or revised product offerings.

6) Monthly profit trends and month-over-month comparisons.

I first create a new column called Month_trend by extracting the month number from the New_date column. Then, I replace those numeric month values with their full month names using recode(). This way, my Month_trend column is easier to understand because it shows names like "January" instead of just numbers.

```
orders<-orders %>%
      mutate(Month_trend= month(New_date))



Monthly_Trend<-orders  %>%
  mutate(
    ## Convert numeric month codes to month names
    Month_trend=recode(Month_trend, "1"="January",
      "2"="February",
      "3"="March",
      "4"="April",
```

```r
      "5"="May",
      "6"="June",
      "7"="July",
      "8"="August",
      "9"="September",
      "10"="October",
      "11"="November",
      "12"="December"),
      # Set factor levels to order months chronologically for plotting
      Month_trend=factor(Month_trend,
        levels = c("January",
                   "February",
                   "March",
                   "April",
                   "May",
                   "June",
                   "July",
                   "August",
                   "September",
                   "October",
                   "November",
                   "December"))) %>%
          group_by(Month_trend) %>% # Group data by Month_trend
          summarise(Monthly_Profit= sum(profit))# Calculate total profit per month

kable(Monthly_Trend) # View summarized monthly profit data
```

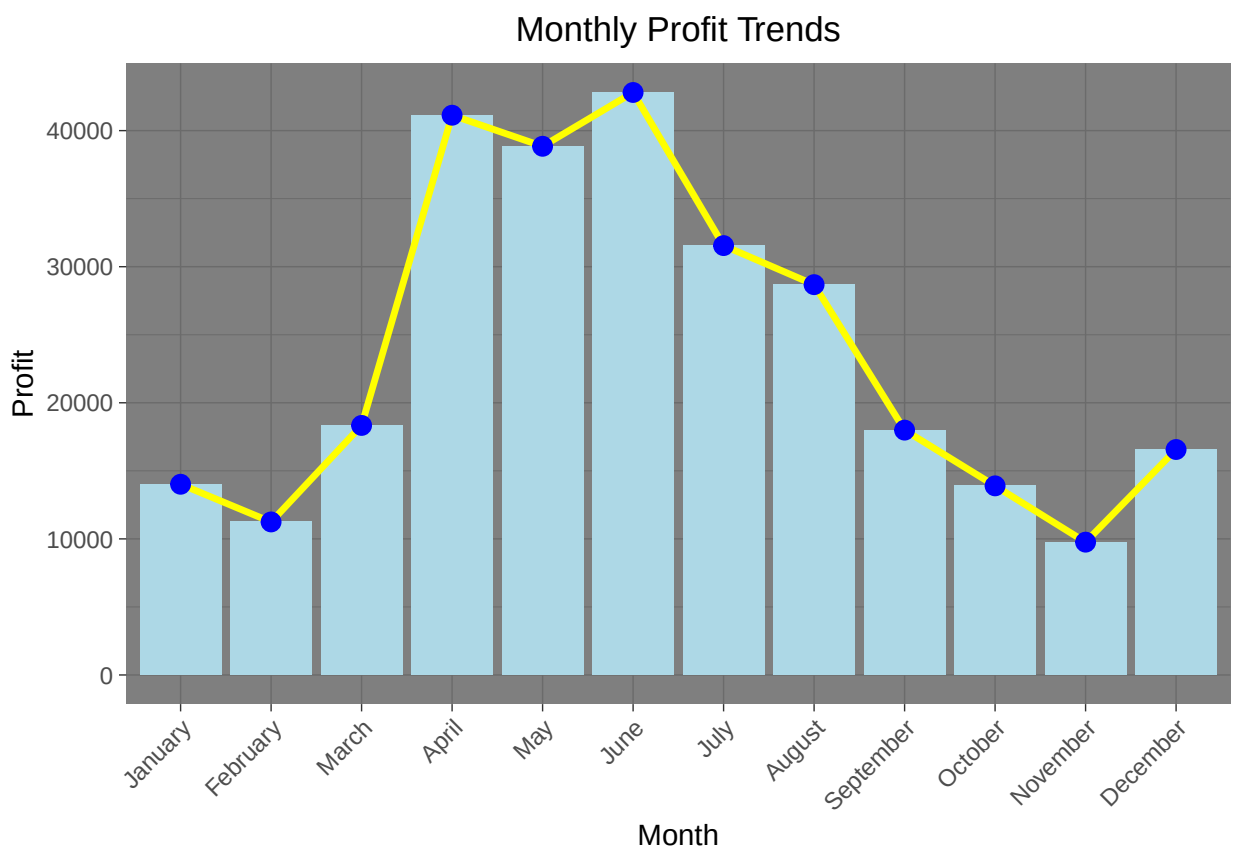
Month_trend	Monthly_Profit
January	14013.52
February	11244.50
March	18335.76
April	41131.00
May	38847.24
June	42802.26
July	31550.40
August	28681.36
September	17992.00
October	13895.44
November	9760.52
December	16567.86

Monthly Profit Trend Graphic

```r
ggplot(Monthly_Trend, aes(x = Month_trend, y = Monthly_Profit)) +
  geom_bar(stat = "identity", fill = "lightblue") +
  geom_line(aes(group = 1), color = "yellow", size = 1.2) +
  geom_point(color = "blue", size = 3) +
  theme_dark()  +
    theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(hjust = 0.5)
```

```r
  ) +
  labs(
    title="Monthly Profit Trends",
    x= "Month",
    y="Profit"
  )
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



- June has the highest profit ($42,802), marking a mid-year sales peak.
- April ($41,131) and May ($38,847) also show strong spring profits.
- January ($14,014) and February ($11,245) have low profits, likely due to post-holiday slowdowns.
- November has the lowest profit ($9,761), possibly indicating inventory or sales issues despite the holiday season.
- Profit fluctuates month-to-month, with sharp increases from March to April and declines after July.
- July ($31,550) dips compared to June, suggesting a mid-summer slowdown.

- September ($17,992) and October ($13,895) are relatively low, reflecting seasonal variation or challenges.
- December ($16,568) rebounds after November but stays below spring/summer levels.
- Overall, there is clear seasonality with peaks in late spring/early summer and dips in late fall/early year.

7) Monthly Revenue trends and month-over-month comparisons.

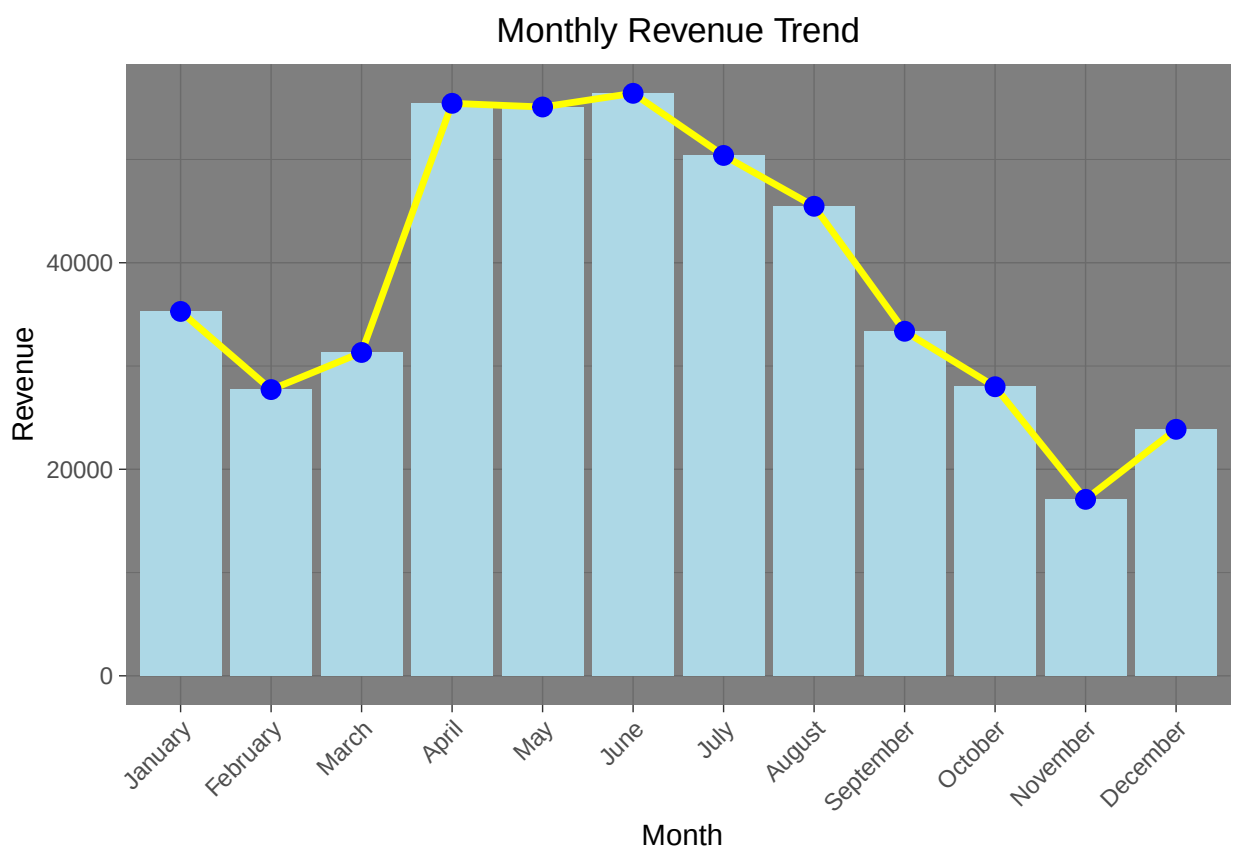
```
Revenue_Trend<-orders %>%
  mutate(
    ## Convert numeric month codes to month names
    Month_trend =recode( Month_trend,  "1"="January",
                                       "2"="February",
                                       "3"="March",
                                       "4"="April",
                                       "5"="May",
                                       "6"="June",
                                       "7"="July",
                                       "8"="August",
                                       "9"="September",
                                       "10"="October",
                                       "11"="November",
                                       "12"="December"),
    # Set factor levels to order months chronologically for plotting
                Month_trend=factor(Month_trend, levels= c("January",
                                                          "February",
                                                          "March",
                                                          "April",
                                                          "May",
                                                          "June",
                                                          "July",
                                                          "August",
                                                          "September",
                                                          "October",
                                                          "November",
                                                          "December"))) %>%
                  group_by(Month_trend) %>% # Group data by Month_trend
                  summarise(Revenue= sum(revenue))# Calculate total revenue per month

Revenue_Trend
```

```
## # A tibble: 12 x 2
##    Month_trend Revenue
##    <fct>         <dbl>
##  1 January      35283.
##  2 February     27718.
##  3 March        31311.
##  4 April        55438.
##  5 May          55082.
##  6 June         56407.
##  7 July         50390.
```

```
##  8 August       45469.
##  9 September    33367.
## 10 October      27995.
## 11 November     17088.
## 12 December     23870.
```

Monthly Revenue Trend Graphic

```
ggplot(Revenue_Trend, aes(x=Month_trend,y=Revenue))+
      geom_bar(stat = "identity", fill="lightblue") +
      geom_line(aes(group=1), color="yellow", size=1.2) +
      geom_point(color= "blue", size=3) +theme_dark()+
      theme( axis.text.x = element_text(angle= 45, hjust=1),
             plot.title = element_text (hjust = 0.5))+
       labs( title= "Monthly Revenue Trend",
        x= " Month",
        y="Revenue")
```



- June had the highest revenue ($56,407), marking peak business in early summer.

- April ($55,438) and May ($55,082) showed strong spring revenue.
- July ($50,390) and August ($45,469) maintained solid summer performance.
- November recorded the lowest revenue ($17,088), unusually low despite events like Black Friday.
- December revenue ($23,870) improved from November but stayed below yearly average.

- January to March had modest revenues ($27K–$35K), indicating a slow start.
- Sharp increase from March to April suggests seasonal growth or successful campaigns.
- Revenue declined gradually from September ($33,367) to October ($27,995), showing post-summer slowdown.
- Overall, the data reveals clear seasonality with strong spring and early summer performance.

Executive Summary

The business demonstrates strong profitability with an average profit margin of 62%, highlighting sports such as soccer and basketball for their volume and profit margins.

Customer ratings do not show a direct correlation with profitability; however, low ratings indicate potential areas for improvement.

Geographically, states like Utah and Massachusetts stand out for margin efficiency, while high-volume states such as California, Texas, and Florida dominate in total revenue and profit.

Seasonality is evident, with peaks in spring and summer (April to June) and notable declines in winter and fall, especially in November.

It is recommended to focus marketing campaigns during peak months and investigate the causes of low profitability in underperforming states and months.